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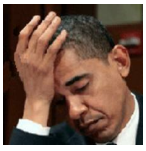
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## Features

May/June 2011 The Information Sage

Meet Edward Tufte, the graphics guru to the power elite who is revolutionizing how we see data.

By Joshua Yaffa

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Illustrated by Merchant for the Brunswick Review

One day in the spring of 2009, Edward Tufte, the statistician and graphic design theorist, took the train from his home in Cheshire, Connecticut, to Washington, D.C., for a meeting with a few members of the Obama administration. A few weeks earlier, he had received a phone call from Earl Devaney, a former inspector general in the Department of the Interior, who is best known for leading that agency's investigation of the lobbyist Jack Abramoff. Devaney had recently been appointed head of the Recovery Act Accountability and Transparency Board, the body created by the Obama administration to keep track of the $780 billion in federal stimulus money that has spread out across the country.

Whereas Devaney once led a team of professional investigators responsible for sniffing out waste, fraud, and abuse, he was now faced with a rather different, but related, task: designing a Web site. In the stimulus bill, Congress had called for the creation of "user-friendly visual presentations" of data that would allow the American public to watch over the disbursement of the giant funding package. This wasn't exactly familiar territory for Devaney, a career lawman. Perhaps Tufte could offer some advice?

And so, that April, in an office building blocks from the White House, Tufte spent a few hours with Devaney looking at sketches of some of the displays the board was preparing. Devaney showed Tufte a prototype of Recovery.gov, the site that catalogs all the projects funded with federal stimulus money around the country. Thinking about it now, Devaney remembers that the proposed pages were full of "classic Web site gobbledygook, with lots of simple pie charts and bar graphs." Tufte took one look at the Web site mockups that the board's designer had prepared and pronounced them "intellectually impoverished."

It was a classic Tufte moment: a spontaneous and undiplomatic assessment that immediately struck everyone in the room, even the designer himself, as undeniably true. The site would get a wholesale redesign. The model, as Tufte explained it, should be the Web site of a major newspaper, with Devaney and his staff as reporters and editors. "I told them that it isn't an annual report," Tufte told me later. "It shouldn't look stylish or slick. It's about facts." As Tufte and Devaney talked, a number of staffers gathered in the hall, waiting for the meeting to finish. "The guys from the IT department had lined up outside my door to shake his hand and say they met the guy," Devaney remembers.

Edward Tufte occupies a revered and solitary place in the world of graphic design. Over the last three decades, he has become a kind of oracle in the growing field of data visualization—the practice of taking the

sprawling, messy universe of information that makes up the quantitative backbone of everyday life and turning it into an understandable story. His four books on the subject have sold almost two million copies, and in his crusade against euphemism and gloss, he casts a shadow over the world of graphs and charts similar to the specter of George Orwell over essay and argument.

Tufte is a philosopher king who reigns over his field largely because he invented it. For years, graphic designers were regarded as decorators, whose primary job was to dress up facts with pretty pictures. Tufte introduced a reverence for math and science to the discipline and, in turn, codified the rules that would create a new one, which has come to be called, alternatively, information design or analytical design. His is often the authoritative word on what makes a good chart or graph, and over the years his influence has changed the way places like the *Wall Street Journal* and NASA display data.

In policy circles, he has exerted a quiet but profound influence on those seeking to harness the terabytes of data flowing out of government offices. In recent years, several large American cities, including New York, Oakland, and Washington, D.C., have opened up entire universes of municipal statistics, giving birth to a cottage industry of programs and applications that chart everything from the best commuting routes to block-by-block crime patterns. And under the Obama administration's Open Government Directive of 2009, the federal government has been releasing scores of downloadable data sets. In the public realm, data has never been more ubiquitous—or more valuable to those who know how to use it. "If you display information the right way, anybody can be an analyst," Tufte once told me. "Anybody can be an investigator."

Tufte is equal parts historian, critic, and traveling revival preacher. For a few days each month, he goes on the road to teach a course called "Presenting Data and Information" in hotel ballrooms and convention centers. One afternoon, shortly after his meeting with Devaney, Tufte was teaching his course in the downstairs ballroom at the Marriott hotel in Seattle, Washington. There were 400 people in the crowd. Tufte, who is sixty-nine, and has a thinning slash of silver hair and a pair of wire-rimmed glasses that hang off his nose, was standing at the front of the room. He was wearing a wireless microphone, and held a copy of a small map above his head.

"This," he said, "is *War and Peace* as told by a visual Tolstoy." The map is about the size of a car window, and follows the French invasion of Russia in 1812. It was drawn in 1869 by a French engineer named Charles Joseph Minard. On the left of the map, on the banks of the Niemen River, near Kovno in modern-day Lithuania, a horizontal tan stripe represents the initial invasion force of 420,000 French soldiers. As they march east, toward Moscow—to the right, on the map—they begin to die, and the stripe narrows.



The map itself is elegant and restrained, but it tells the story of a sprawling, bloody horror: cold and hunger begin to finish off whatever French soldiers the Russians haven't killed in battle. As the French army retreats, the tan line turns black and doubles back on itself to the left, or to the west, away from Moscow. A series of thin gray lines intersect the path of the army, showing the winter's cruel temperatures. On November 9 it is 9 degrees below freezing. On November 14, it is 21 degrees below.

Then, on the 28th of November, a catastrophe: in a rush to cross the Berezina River, half of the retreating army, 22,000 men, drowns in the river's icy waters. The black line, already thinned to a fraction of its initial size, abruptly reduces by half. Finally, in late December, six months after they set off, the surviving French soldiers cross back over the Niemen River. The map shows their number: 10,000 men. Ultimately, only one in forty-two soldiers survives the doomed campaign.

Tufte pointed to the far left of the map, where the tan and black lines intersect. "And it is there," he said, "at the beginning and at the end of the campaign, where we have a small but poignant example of the first grand

principle of analytical design": above all else, always show comparisons.

Over the years, in his books and courses, Tufte has called the Minard map perhaps the best statistical graphic ever made, and his endorsement has turned this obscure historical artifact into something of a cult object. Tufte's long and discursive speech on the map serves as the dramatic climax of his courses, the equivalent of an old rock band playing its greatest hit on tour.

He usually does his explication of the map just before lunch. In Seattle, we were about to break, and he was finishing up. His voice quickened. "Minard made this because he hated war," he said. The map wasn't about Napoleon, the war's surviving hero, who is mentioned nowhere on the page, Tufte explained, but of the quiet, anonymous misery of tens of thousands of French soldiers. "This was meant as an antiwar poster."

The man sitting behind me, a middle-aged software engineer with a laptop, let out a puff of air. "Wow."

The course finished at five that evening. I walked outside to the hotel's parking lot, where a white stretch limousine was parked near the entrance. Two men in loose black suits were leaning against the hood, and they introduced themselves as the limo's drivers, Vlad and Vlad. A minute later, Tufte came out, his face tired and hair wet. "These shows can really wipe me out," he said. Behind him walked two roadies who travel with him from city to city, helping to set up the equipment for each class and handling registration. We all got in the limo, pulled onto Interstate 5, and headed toward Portland, where Tufte would be teaching the course again two days later.

Tufte, who taught political science and statistics at Yale from 1977 to 1999, first gave his six-hour course in 1993. Since then, more than 190,000 people have attended. The day costs $380, which includes Tufte's four books, which he self-publishes under the name Graphics Press. He is a very wealthy man. In his carriage and attitude, Tufte appears most comfortable still thinking of himself as an academic: at his courses, many people in the audience call him "Professor," and during breaks and lunch he answers questions and signs books as part of "office hours." (More than a decade after leaving the university, he still uses his yale.edu email address.) But he is not beholden to any institution, nor to any product except his own, of which he has become a highly refined salesman.

Tufte's rise mirrors that of information itself, which, since the dawn of the computer age, has infiltrated every aspect of modern life. If the production of data and its graphic representation was once a specialized trade, it is now accessible to nearly everyone. Information is all around us, but so is what Richard Saul Wurman, an architect who, along with Tufte, shaped the field of information design in the 1970s and '80s, calls "non-information," the flotsam and noise that are the by-products of a hyperactively quantitative culture.

The modern history of information design has its roots in the Bauhaus, an arts institute founded in 1919 in Weimar, Germany. Later, in postwar America, designers like Nigel Holmes at *Time* magazine and George Rorick, who invented the modern weather map at *USA Today*, began to draw playful graphics that boiled down a question to a few numbers and then presented them in a colorful package. They were inviting and simple drawings, like a cascading stack of oil drums to show the rising price of oil, and were generally more full of good cheer than hard data. Tufte has derided the genre, calling them "chartoons."

For a long time, there was a split between designers who were making graphics for mass consumption and scientists who relied on graphics to communicate their findings. What was meant to be accessible should not look too technical, while for those making complex and substantive points, there was a sort of honor in making impenetrable graphics. "Making something look good was considered treacherous," said Erik Spiekermann, a German typographer and designer who created the typeface used on German railways. "If you make something look pretty it means you're lying. If you want something true it's got to hurt."

In his books, the first of which, *The Visual Display of Quantitative Information*, came out in 1983, Tufte aimed to destroy this myth, largely by relying on historical examples like the Minard map. "Tufte killed the idea that we are afraid of numbers," said Tobias Frere-Jones, a typographer who keeps Tufte's books on a shelf above his desk in downtown New York. "And once you get over that idea, you can't really justify the birthday-party-clown school of data visualization, where you need bright colors and shiny things to convey that the stock market went down this week."

Tufte has coined several terms that have come to define his style, such as "data-ink ratio," the proportion of graphical detail that does not represent statistical information, and "chartjunk," ornamental and often saccharine design flourishes that impede understanding. Jonathan Corum, a former graduate student of Tufte's at Yale who is now the science graphics editor at the *New York Times*, said that using Tufte's principles is a way of respecting the reader's intelligence. "I think of it like I'm starting with this rough, unpolished stone," he said. "So how do you cut the perfect gem?"

Many of Tufte's neologisms have come to carry a sort of Talmudic authority in design firms and newsrooms, though they are also quoted by engineers, financial analysts, consultants, and, perhaps unsurprisingly, political strategists. Karl Rove, senior adviser to former President George W. Bush, attended one of Tufte's courses in Austin in the 1990s and considers himself a "huge" fan. "I must have bought a hundredplus copies" of *The Visual Display of Quantitative Information* for colleagues and friends, he told me. "Particularly in the early days in the White House, someone would come in and show me a presentation full of chartjunk"—in this context, Rove explained, extraneous graphics or, say, bar charts drawn from interlocking dollar signs—"and I would send them off with Tufte's book to do it again."

Tufte may be one of the few theorists—design or otherwise—lauded in both the Bush and Obama administrations. After their first meeting on Pennsylvania Avenue, Earl Devaney sent Tufte's resume to the White House, leading to his appointment last March as an official adviser to the Recovery Transparency and Accountability Board. But his influence in the administration has reverberated beyond that office. Peter Orszag, who left his post as head of Obama's Office of Management and Budget last year and then took a job as vice chairman of global banking at Citigroup, spoke to me recently in his office on the thirty-ninth floor of the bank's headquarters in lower Manhattan. He was first given a copy of Tufte's *Visual Display of Quantitative Information* as a college graduation present. As Orszag explained, the challenge of designing graphics to represent the nearly infinite stream of data coming out of government offices is to be compelling but not to oversimplify. The description sounded like many of the maxims I heard on tour with Tufte. Indeed, at the end of our conversation, Orszag told me that "Tufte's spirit" has "been taken to heart by an entire generation of data analysts."

Before his work with Devaney, Tufte's most public engagement with the federal government came in the form of a scathing crusade against its endemic use of Microsoft PowerPoint. In January 2003, the space shuttle *Columbia* took off from Kennedy Space Center. Seconds into its flight, a piece of foam insulation broke off from the shuttle's fuel tank and punctured its left wing. For days, as the shuttle orbited the earth, engineers at NASA and Boeing debated how serious the hole might be. Much of this communication was done in the form of PowerPoint, a software program that Tufte says is constricting and obfuscating and "turns information into a sales pitch."

After reviewing several PowerPoint presentations, NASA officials decided to proceed with the shuttle's return flight. On February 1, *Columbia* burned up as it reentered the earth's atmosphere, killing all seven crewmembers on board. Tufte has called PowerPoint a "coconspirator" in the shuttle disaster.

Tufte dissected NASA's PowerPoint slides on his Web site, showing that the program didn't allow engineers to write in scientific notation and replaced complex quantitative measurement with imprecise words like "significant." He then published a twenty-eight-page essay called "The Cognitive Style of PowerPoint," in

which he analyzed hundreds of existing PowerPoint slides and showed that the statistical graphics used in PowerPoint presentations show an average of twelve numbers each, which, in Tufte's analysis, ranks it below every major world publication except for *Pravda*. The low information density of PowerPoint is "approaching dementia," he wrote.

Later that year, in August 2003, the Columbia Accident Investigation Board issued its own report on the shuttle accident. On page 191, in a section called "Engineering By Viewgraphs," the board cited Tufte's research, and wrote that it "views the endemic use of PowerPoint slides" as "an illustration of the problematic methods of technical communication at NASA."

While PowerPoint still takes a rather severe lashing at Tufte's courses, he says he has largely moved on from the debate. "It's a public and corporate disaster for Microsoft," he said. "But it's a trivial issue to me." The general backlash against PowerPoint has only grown, however. In 2009, Thomas X. Hammes, a retired Marine Corps colonel who is now a fellow at the National Defense University, wrote a widely circulated essay in *Armed Forces Journal* that argues that PowerPoint has had a "toxic effect" on the culture of debate and discussion within the U.S. military, claiming that "senior decision-makers are making more decisions with less preparation and less time for thought."

As Hammes explains it, the reliance on PowerPoint often means that battle orders are rendered in incomplete, often unclear sentences and maps are squashed and stripped of meaningful detail, leaving essential battlefield questions of geography dangerously unclear. The details are classified, but Hammes told me that he has seen war plans for the Korean peninsula prepared in PowerPoint in which massive terrain issues were completely glossed over. On the whole, Hammes told me, the rise of PowerPoint in the military has made the decision-making process less intellectually active. And Tufte, he added, "is the master on this whole thing."

The spread of technology over recent decades has fostered a comfort with data, and an appetite for more of it, in an ever-increasing portion of the general public. This cultural shift was especially visible in the dense data displays that appeared in newspapers, and especially online, in the run-up to the 2008 presidential election. Here, too, Tufte's influence was evident. Nate Silver, who runs the political Web site FiveThirtyEight, now part of the *New York Times*, uses many of Tufte's maxims in the site's design. Silver told me that he tries to keep the "data-ink" ratio of his current site very high, meaning most of the pixels on the screen show actual numbers or data points; he also thinks of the site's design in terms of "small multiples," another Tufte neologism that refers to a series of related numbers that reveal subtle differences over time. "Tufte treats data like good writing," he said. "You have a certain thought—how clearly and beautifully are you conveying it?"

Good design, then, is not about making dull numbers somehow become magically exhilarating, it is about picking the right numbers in the first place. "It's about data that matters to you," said Dona Wong, who was a student of Tufte's at Yale in the 1980s and later the graphics director at the *Wall Street Journal*.

This idea is especially relevant for how the federal government measures and displays the impact of the stimulus. What are the most informative comparisons? And what, for lack of a better word, is simply chartjunk? Tufte first suggested that the recovery board rethink how the projects are ranked. Until now, they have been sorted by location, with the idea that any person could look up projects in his or her county or zip code. "But what's it matter where the project is? We already know where Texas and California are," Tufte said. Geographic space helps little in this context. "Always order things substantively," Tufte explained. He would like to bring in all other kinds of data sources to add context to the stimulus and track its actual effects. Soon, if Tufte has his way, Recovery. gov will display analytical graphics that measure funded projects alongside metrics like infant mortality, traffic flows, and education. After all, as he told me, "understanding doesn't come from project descriptions."

Someday, the Web site might also use what has proven to be one of Tufte's more popular inventions. In his

2007 book *Beautiful Evidence*, Tufte introduced what he called "sparklines," numerically dense, word-size graphics that show variation over time. They have since appeared on the financial pages of Yahoo! and the sports section of the *New York Times*. (As an example, the fluctuations of the Dow Jones Industrial Average over the course of February and March look like this:  . The dramatic dip in the figure represents the March 16th panic over the nuclear disaster in Japan.)

In November 2009, Tufte's old bête noire, Microsoft, attempted to patent sparklines for use within the spreadsheets that will appear in the latest version of Excel. The notion that the company could patent an innovation that Tufte has championed for several years struck him as "a great lark," and a sign of "characteristic overreaching" on the part of Microsoft. But for a man who clearly takes no small satisfaction in seeing his ideas adopted, even by nefarious software conglomerates, it wasn't an entirely awful development. "I'm happy," Tufte told me. "Sparklines are everywhere else, so why not in Excel too?"

The underlying philosophy behind sparklines—and, really, all of Tufte's work—is that data, when presented elegantly and with respect, is not confounding but clarifying. (Click here see an example). "There is no such thing as information overload," Tufte says at the start of his courses. "Only bad design." (In Portland, this last line received some applause, as well as a cry of "God, yes!")

As Richard Grefe, the executive director of the American Institute of Graphic Artists, which awarded Tufte its highest medal in 2004, explained, Tufte has shifted how designers approach the job of turning information into understanding. "It's not about making the complex simple," Grefe told me. "It's about making the complex clear."

Tufte was born in 1942 in Kansas City, Missouri. His mother graduated high school at the age of thirteen, and four years later she became the first female reporter at the *Omaha World-Herald*. His father was an engineer, and Tufte remembers his parents' marriage as the first sign that "words and numbers belong together."

He studied statistics at Stanford and then went on to get a doctorate in political science at Yale. In 1967, he took a job teaching at Princeton. While there, Tufte was asked to give a course on statistics to a group of visiting journalists and, in looking for examples to include in the course packet, quickly became dissatisfied with the available primers on how to represent data. They were either too shallow and unserious or hopelessly arcane. He began to write up some ideas of his own.

A few years later, Tufte moved to Yale. He became friendly with Inge Druckrey, a German-born designer and teacher who had studied in the 1960s at the Basel School of Design in Switzerland, then an incubator for modernist style. The two would talk about design theory, and Tufte would visit Druckrey's classes to critique student work. Before long, the two began dating.

Soon, Tufte's notes on information design had grown into a book-length manuscript. He showed it around to publishers, who insisted on redesigning many pages in the book, and imagined it as a niche title, only worth printing a couple thousand copies. Frustrated, Tufte took out a second mortgage on his home at 18 percent interest to print the book himself. He spent most of the next summer with a book designer named Howard Gralla. The two of them sat side by side in Gralla's apartment, eating bagels and rearranging text so words and images would be woven together on the page. "Self-publishing," Tufte told me, "allowed for an incredible, bizarre fussiness."

*The Visual Display of Quantitative Information* came out in April 1983. To save costs, Tufte told the printer to bind only half of the initial print run of 5,000 copies. The book is now in its twentieth printing, and is one of the most successful self-published books of all time.

As one of his close friends, the psychologist and author Paul Ekman, told me, Tufte is a "profoundly

unconventional fellow." During those years in New Haven, Tufte would leave daily memos for his staff in a shoe box. One day, when he had been dating Druckrey for some time, his office manager came in to work, opened the shoe box, and saw a piece of torn paper that said, "Find out how to get married." Druckrey was taken with Tufte's intensity. "He was endlessly curious," she told me. "He has this special eye and judgment, and was beaming with ideas." In 1985, they married.

If Tufte's first book was a critique, his second was a manifesto. *Envisioning Information*, published in 1990, implored readers to think of information design as a discipline that encompassed far more than the charts, tables, and other purely quantitative forms that had traditionally dominated the field. Graphics aren't just useful for displaying numbers, in other words, but for clarifying just about anything one person is trying to tell someone else. The book opens with a print of a visitor's guide to the Ise shrine in Japan and ends around 120 pages later with Galileo Galilei's drawing of the rings of Saturn from 1613.

Galileo is Tufte's greatest inspiration as a designer. "He is the best visual thinker ever, the best graphic designer ever," Tufte once told me. During his courses, he pulls out an original copy from 1613 of Galileo's *Istoria e Dimostrazioni Intorno Alle Macchie Solari* from a duffel bag, and opens it to a page with a passage describing Saturn and its rings. In the book, Galileo uses words and drawings laid out next to one another to show how the planet appeared on different nights. ("The shape of Saturn is thus  as shown by perfect vision and perfect instruments," Galileo writes, "but appears thus  where perfection is lacking.") It is the same idea that showed up almost 400 years later in Tufte's sparklines.

"Galileo faced the same display issues, and I want to see how he solved them," Tufte said one day in his library at his home in Connecticut. In doing research and making prints for his books, Tufte has insisted on working only from original documents, which has made him a rather prolific collector. In one visit to his library, Tufte showed me a Victorian book diagramming popular dance steps of the time, a sketchbook made by Pablo Picasso during his Paris years, and a copy of the *Hypnerotomachia*, a dreamlike love story that was printed in Venice in 1499, which Tufte bought at auction for $385,000. On another shelf was a first edition of *Starry Messenger*, Galileo's book from 1610 that was the first to rely on observations from a telescope (which Tufte has since sold). "It is a reminder to be better, to work harder," Tufte said.

Tufte's books feel unattached from any time or place. They are anachronistic in their images, styling, and soft-beige paper, as if dropped into the present from a vague and altogether more genteel past. A woman in the audience at his course in Seattle said that when she first saw Tufte's books she thought they were college textbooks long out of print; another man told me his friends at work had assumed "that guy named Edward Tufte" was British and died years ago.

After the publication of *Envisioning Information*, Tufte decided, he told me, "to be indifferent to culture or history or time." He became increasingly consumed with what he calls "forever knowledge," or the idea that design is meant to guide fundamental cognitive tasks and therefore is rooted in principles that apply regardless of the material being displayed and the technology used to produce it. As Tufte explains it, basic human cognitive questions are universal, which means that design questions should be universal too. "I purposely don't write books with names like *How to Design a Web Site* or *How to Make a Presentation*," he told me.

This attitude puts him in opposition, at least in his own mind, to much of the contemporary design world. As Tufte sees it, graphic design has become a tragic field, a rich and storied craft knowledge that has been taken out of the realm of "nonfiction," as he calls it, and into that of "fiction," or marketing and propaganda. He told me several times of his contempt for "commercial art," the graphic design that is "part of a fashion and a style and will be different someday." Most designers, he said, want to do something new each time. "But I'm interested in the solved problem," he said. "I'm interested in high art and real science."

Some designers have questioned whether Tufte's reverence for elegance and accuracy can verge on dogmatism, with too little consideration of context or audience. "The world is not filled with professional statisticians," said Donald Norman, the codirector of the Segal Design Institute at Northwestern University and the author of *The Design of Everyday Things*. "Many of us would like a quick glance just to get a good idea of something. If a graph is made easier to understand by such irrelevancies as a pile of oil cans or cars, then I say all the better." (Tufte deflects this criticism by pointing out that Norman has been a paid consultant to Microsoft; Norman says his consulting work has nothing to do with his own thinking and writing.)

But when Tufte feels sure of something—which is to say, quite often—he can have little interest in entertaining criticism. "There are some people who have reached a certain position in life and will be amused and interested in a contrary opinion, and have a scholarly banter about it," said Christopher Pullman, who taught in the design program at Yale on and off since the 1960s. "But that's not Tufte."

After decades of battling over the proper use of two dimensional space, Tufte has turned much of his attention in recent years to a new undertaking: outdoor sculpture. It is perhaps no coincidence that he called his first piece, made in 1997, *Escaping Flatland*—a name that not only describes the work's bending waves of stainless steel that jut out from the ground, but also evokes a sort of personal emancipation. "So much of what I did before was a war against stupidity," he told me not long ago. "With sculpture it's just about private pleasure."

A couple of years ago, the Aldrich Museum of Contemporary Art, not far from Tufte's house in Connecticut, invited him to put on a solo exhibition of his work, which mainly consists of abstract pieces built at extremely large scale. "On to the fourth career," Tufte joked with friends. The exhibition ran until the spring of 2010; then, in April, Tufte signed a lease for an empty gallery space on West Twentieth Street in New York's Chelsea arts district to continue showing his work. One day over the summer, Tufte was in the gallery making some final adjustments. One long, crane-necked metal sculpture wasn't quite catching the light, and a piece that looks like a cartoonish lunar module had to be shifted so it was more visible through the gallery's picture windows.

Tufte had just returned from Washington. Pro bono consulting for government has meant a rather noticeable lifestyle change—in the past, when Tufte traveled to Washington to teach his courses, he stayed at the Hay-Adams, a stately hotel on Lafayette Square, just across from the White House. Now the government puts him up at the Crowne Plaza across the river in Arlington, Virginia, which, as Tufte described it, seems to be "largely populated by fifteen-year-olds from North Dakota." He showed me his GSA Smart Pay Travel Card, the government-issued credit card that federal employees and contractors use to pay for expenses. "Soldiers have them, too," he said, his voice buoyant and proud. He laughed. "But it comes with stern prohibitions about the minibar."

So far, Tufte's most visible contribution to Recovery.gov is a map that he designed, which shows the rollout of 101,236 stimulus projects between February 2009 and December 2010 as a proliferating series of white lights overlaid on top of a nighttime map of the United States. Tufte has been pushing Devaney and the board to do more with the data it collects and faster. But acquainting Tufte with the slow, procedural pace of bureaucracy has been the greatest challenge, Devaney told me. "Tufte drives race cars," Devaney said, "and most people in Washington drive tanks."

We took a walk around the gallery, stepping past a large, bemused-looking fish cast in divoted shiny metal. Tufte started telling me about the official appointment letter that he was issued by the White House. It is an ornate and formal document, with the president's signature in the lower left corner. Soon after receiving it, Tufte took it to a frame shop near his house in Connecticut. The framer came back with a proposal. It's too square, he told Tufte, not at all the sort of "golden rectangle" that artists and designers tend to favor. Perhaps he could cut a few inches here and there to make it look more presentable in the frame? Tufte politely

declined. “That’s the very reason I’m here!” he told me, laughing. “To fight against decoration replacing precious substance.”

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Joshua Yaffa is an associate editor at *Foreign Affairs*.
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### Comments

Post a comment

- **Dona Wong** on May 11, 2011 3:12 PM:

  Thanks for a wonderful article.

  Best regards,
  Dona Wong
  Author, The Wall Street Journal Guide to Information Graphics

- **tom o'neill** on May 11, 2011 9:18 PM:

  I was fortunate to be a graduate student of Tufte's at Princeton's Woodrow Wilson School in 1968-70. As a teacher he was mesmerizing and enormously effective, with the ability not only to teach liberal arts majors to master regression analysis (done by hand on a mechanical Frieden calculator) but to present the results clearly and graphically. He taught a valuable lesson in his comment on one long paper I did that tried to predict the outbreak of civil disturbance in countries around the world based on a disequilibrium between the nation's economic versus its civic development. His scrawled comment on the 100 page paper, which began with an explanation of the methodology "It's page 36 and nothing has happened yet." I still have that paper, and I treasure it.

- **Doug Stickler** on May 12, 2011 2:58 PM:

  I had the pleasure of attending one of Tufte's workshops several years ago in silicon valley. He presented very well the universal nature of design across all information domains. I found it to be refreshing and valuable as are his books which I have utilized extensively as well as his website. But the most remarkable thing for me was discovering how engaged and bright he is when I met him during 'office hours'. He immediately started asking me questions. He wanted to know what I did briefly described(healthcare analytics) and then proceeded to jump to the heart of some complex problems I hadn't even considered much less articulated. Then he literally pointed me to specific sections in his books. It was a very information rich three minutes! I'm not much of a groupie but I do treasure his comments and notes in my books. This article captures much what makes Tufte special.

- **John Ashton** on May 13, 2011 4:57 AM:

  First saw all this stuff a few years ago and it was "gee wow! great! must do this with my data" and so on.

  But...after a while you realise that it is just pretty. It isn't informative, it doesn't help understanding. It's just pretty and useless.

- **Leanne Tobias** on May 13, 2011 6:32 AM:

  I was lucky to be a student of Professor Tufte's at Princeton's Woodrow Wilson School, shortly before he began teaching at Yale. Professor Tufte taught a data analysis course that blended statistics, linear regression analysis and data presentation techniques. Decades later, I still remember Professor Tufte's maxims and try to follow his advice when using and presenting data. His advice ranged from the simple (clearly label the axes of charts and graphs; never distort data through graphical manipulation) to the more complex (regression coefficients are not meaningful unless they are expressed in units clearly related to the performance of the variables used in the analysis).

  When I was his student, Professor Tufte blended wit, intellectual acuity and a passion for clarity. I'm glad to read that he's better than ever. Kudos to Professor Tufte and to Washington Monthly for a great article.

  Leanne Tobias
  Malachite LLC
  Bethesda, MD

- **inchoate but earnest** on May 13, 2011 8:14 AM:

  Fine piece overall.

  We find a couple of inexplicable bits.

  The first is Mr Ashton's comment. Mr. A, you're not clear what the "it" is you're whining about, but if "it" is graphical depiction of information generally, or Tufte's perspective on such depiction specifically, your comment is the useless article.

  Second,tufte's comment re: the significance of geography in the depiction of project data: The guy's a marvel, but like other humans missteps occasionally, as here, with respect to Recovery.gov:

  > But what's it matter where the project is? We already know where Texas and California are, Tufte said. Geographic space helps little in this context.
  >
  > Always order things substantively, Tufte explained. "

  What it matters is that all politics are local. What it matters is that those "substantive" projects show up in places for a reason - mostly as the results of distributions of political power across the landscape.

  So how do you show that important aspect? In a list? Does Texas appearing above North Carolina above Connecticut above Utah tell me something useful about the politics of their position - as much as their geography might?

  Disagree with geographic map depictions, but ignoring the "locality" of decisionmaking leaves an

important aspect of the data literally out of the picture. Tell us - better, *show* us - how you'd do it differently.

- **bob white** on May 13, 2011 9:28 AM:

  In addition to everything else, Edward Tufte is an extremely nice man. About twenty years ago my German Shepherd Chance managed to eat the dust cover of The Visual Display of Information. Knowing that it was self-published,I wrote to him, explained the situation, and asked if I could buy a new cover.
  Well, not only did he send me the cover without charge, he also included a rawhide bone with a note "for your puppy".
  Some years later, there was an article about him in the New York Times Magazine. In the picture were his two Goldens...all became clear to me.
  If he happens to read this, I thank him again. Probably, he will not notice that I am not using my real name.

- **Steve Macdonald** on May 13, 2011 10:14 AM:

  Fascinating article. Tufte is evidently a genius, about whom I'd known little heretofore. The repeated mentions of graphics depicting the effects of the stimulus do however call to mind the old refrain, "lies, damned lies, and statistics". I do not wish to take any position on the wisdom of the stimulus, but rather to point out that such economic actions are highly debatable and by no means "proven" except in the minds of true believers, for whom no diagrams are needed in any event. It sounded like the graphics in question were to some extent accepting stimulus spending as "gospel" (even though for example similar policy measures were roundly rejected throughout Europe). This does not communicate truth, but rather partisan political propaganda.

  Otherwise I believe Tufte is a genius as stated. I am simply wary of anyone claiming to have a handle on truth or even transparency when it comes to something of near infinite complexity such as the US economy.

- **Don Nadeau** on May 13, 2011 5:49 PM:

  @Steve Macdonald: Gee I dunno, Steve, the bridges around my city were falling apart until a stimulus package set a bunch of design and construction folks (white and blue collar for you visual types) to work fixing them, so the capitalists would feel better routing their truckers (um gold collar and gray collar?) trucks full of product over them, rather than the time-consuming, more expensive, and more energy-guzzling alternative long way around. The oldest bridge in the nation is even getting fixed, huzzah!

- **Don Nadeau** on May 13, 2011 6:45 PM:

  BTW, the antiquated Reaumur degrees quoted in the article are centered (zeroed) on the freezing point of water. But:

  1* Reaumur = 1.25* Celsius = 2.25* Fahrenheit.

  (The * is my symbol for "degree". Equation source: our friendly neighborhood Wikipedia)

  So the temperatures that the Napoleonic troops endured were, respectively (help me graph this, Professor Tufte!):

- 9 and -21* R
-11 and -26* C
+12 and -15* F

- **Sandio** on May 13, 2011 11:16 PM:

  Let's hear about the cohort fixing trick that made the HIV/AIDS regression. Mathew Edlund on HuffPo Healthy Living points out that you can find a marker for any desired effect, and that's not a fair predictor. This ruse can be reversed: AIDS is not a well-defined condition, and now means only the effect of HIV! Meanwhile the outcome depends on the underlying health of the group chosen...
  These are questions of real contemporary interest: much of the social networking buzz is about tagging along (!) with success. So is Obama spending along with recovery or investing in a real future?

- **Jon H** on May 14, 2011 9:31 PM:

  I *wish* I could catch the train in Cheshire! You have to go south to New Haven first, because it only runs along the coast.

- **Christopher Dillon** on May 16, 2011 2:21 AM:

  "Self-publishing," Tufte told me, "allowed for an incredible, bizarre fussiness."

  Self-publishing also allowed Tufte to produce books that are beautiful as well as useful, something that is very difficult with commercial publishers.

- **Scott Ryburn** on May 16, 2011 2:07 PM:

  I, too, was privileged to attend a Tufte seminar. While his message is so sensible that it is easy to grasp, it is much tougher to execute. I still struggle to apply and refine what I've learned to my profession, financial analysis. It is also not always an easy sell in the business world, where everyone expects "a few slides" to tell the story.

  As a side note to Steve Macdonald regarding the stimulus approach being "roundly rejected throughout Europe," I think it's fair to say that much of what we called "stimulus" was already present in European social safety nets. That damnable old-world socialism!

- **curtis** on May 17, 2011 8:09 PM:

  Those that appreciate what Mr. Tufte is doing might want to read "The Alphabet Versus The Godess" for a perspective on the bigger picture of how information presentation methods might affect human behaviors.

- **Hamish Blair** on May 17, 2011 8:40 PM:

  As an accountant and "frustrated graphic designer" this is perhaps one of the best articles I have ever read. Off to order his book now.

- **El Bosco** on May 18, 2011 1:00 AM:

  Tufte is interesting... if a little (alot) dogmatic. But he should skip the art pretensions. Being good in one field doesn't necessarily mean you're good in another. He's a profoundly average artist, and his last book is the weakest of the four he has produced.

- **FugFuu** on May 18, 2011 9:34 AM:

  Dude jsut looks corrupt as the day is long.

  www.internet-privacy.at.tc

- **Perry Alexander** on May 18, 2011 12:08 PM:

  My learning from a day-long Tufte seminar in the late '90s was an epiphany; a clarification of my passion. I had mistakenly felt what drove me was marketing design. Having worked in that field for 25 years, after that day I knew I was fundamentally in the business of information presentation. Honest marketing design makes use of that skill. Now, graphic design for marketing is one of my avenues to indulge in information presentation, as I aspire to apply the principals of Edward Tufte. I am indebted for the insight, and delighted for the clarity. Great article!

- **Jorge Camoes** on May 20, 2011 6:27 AM:

  Well, I'm a Tufte fan, like almost everyone else in the data visualization community but writing that he is the "...king who reigns over his field largely because he invented it"? That's a bold statement, and far from the truth. In 1967, the French cartographer Jacques Bertin published his "Semiology of graphics" and that's the real data visualization bible. The book was translated into English almost 20 year later and still remains the only consistent theory available.

- **R Path** on May 20, 2011 10:53 AM:

  Nice article; however, one element in the web layout of this story could be improved. You display an image, inline with the text, of 3 graphs showing cancer survival rates, and then have a link to a PDF of a sparkline example. Is the sparkline document supposed to demonstrate an improved display of the same data on the 3 graphs? It's unclear, since the image of 3 graphs cannot be easily enlarged, and there is no reference to that figure in the story itself (or no caption). I find this to be an odd contradiction given that this is an article about Tufte and good design principles...

- **smartalek** on May 21, 2011 4:52 PM:

  Glad I'm not the only one (but disappointed that I appear to be only the second) to note the irony of an article lauding the revolutionary contributions of a visual-data-presentation genius who goes out of his way to stress the importance of comparative data -- an article that has only two graphic examples attached to it, neither of which demonstrates any comparisons of the ostensibly inferior standard techniques against the superior Tufte methods.

  And @ Mr Steve Macdonald (5/13/11 10:14AM):
  Can't speak knowledgeably about the rest of Europe. However, I can point out the instructive -- and directly comparable -- experience of the UK, which famously took the direct opposite of the "stimulus" approach adopted by Pres Obama and the Democratic Congressional majority of the time -- and did what American conservatives in general, and Republicans in particular (with whom it seems Mr Macdonald would stand), were urging for us then (as they are now): they adopted austerity measures, significantly slashing governmental spending.
  Since doing so, their economy has suffered contraction at, IIRC, a rate of about 3-4%. In the same time-frame, and starting at about 6 months after the passage of the stimulus he derides (exactly when one would expect to see the impact), the US has enjoyed growth in the 1-3% range.
  The internet apparently really loves irony. Mr Macdonald is absolutely correct in concluding, "[t]his

does not communicate truth, but rather partisan political propaganda."
But of course it's *his* side of the politico-economic debates that is engaging in what he excoriates.

- **Chad Whitacre** on May 22, 2011 10:12 PM:

  I attended Tufte's course last Monday, and have since been preoccupied with a nagging sense that something about him is off. My unfortunate conclusion: after decades building up a powerful brand, he's now using that brand to play the art market for his own personal gain. This is ironic, because one of his main points is that commercial actors are inherently less credible than scientists and others who act in the public interest. What if he invested his capital and clout in a non-profit that truly acted in the public interest? I'd have a lot more respect for him. As it is, his design principles are wonderful content buried in the chartjunk of his brand.

- **Chad Whitacre** on May 24, 2011 1:42 PM:

  We need to shift the conversation around Tufte from his genius brand to the non-profit he should be starting.

- **Cameron** on May 27, 2011 9:49 PM:

  I went to one of his presentations in the mid-90s.

  Minute for minute is was the most intense intellectual afternoon of my life and it changed the way I think about "data" and "information" for ever. Well worth the money.

  And you didn't even mention the visual display of music.

- **MikeH** on June 02, 2011 11:31 AM:

  He lost me when talking about the space shuttle presentation. The 1986 disaster could have been averted if the detailed scientists did not find the need to explain all of the formulas, calculations, and theories to the final decision makers and could have just said, "Space shuttle go boom!"
  Sure, you should not do full analysis or handing out detailed orders on powerpoint or similar tools, but simplicity and imagery are extremely important when it comes time to make a point.
  I am going to look up one or two of Mr. Tufte's books and think it would be interesting to attend one of his classes, but from the article, he does sound overly academic. To reject the simplified weather symbols is absurd. The average person does not need to know the barometric pressure. A presentation allowing people to glance down and know it is going to be 78 and sunny instantaneously is perfect. Furthermore, if I had an employee a project, I do not want them to to show up at my desk and explain all the little details. I hired them because I think they are capable of doing the analysis and making a recommendation. Feed me the recommendation and if I have any questions, I will ask them. If I needed all of the data, I would have taken on the project myself. Maybe it is just the way the article is written, but I do not understand the need to present such a high level of detail at all times.
  I also agree with the question raised by Mr. Norman. Part of visualization is marketing. How do you reach or entice those who do not inherently share a deep interest in a subject? There is no one size fits all model, and there is no one purpose in producing visual depictions. Unfortunately, we live in a time (I'm not sure, it is just this time) when fiction with a simple, catchy message/image can easily trump facts, unless the facts can also be boiled down to a simple, catchy message or image. In many instances, it would be irresponsible to simply dismiss people who buy into the simple, catchy message.

- **Jerome** on June 08, 2011 2:22 PM:

I am a data analyst by both profession and passion. I purchased a copy of Tufte's version of the Minard map direct from Mr. T's website, only to find out that the map does not fit properly in any standard size frame.

Um, do as I say, not as I do, Mr. T?

- **Avallecdollak** on July 08, 2011 2:28 AM:

  hi there great topic you have going here. Utell2011

- **Tom Emmons** on July 11, 2011 11:02 AM:

  I'm all about limiting powerpoint to what it is good at, but what are some viable alternatives? If I am walking a room through of people through ideas (sales pitch, new product,idea, etc) what am I supposed to use?

  I think finding a suitable replacement is much easier said than done.

- **tryecrot** on August 28, 2011 8:06 AM:

  Yes there should realize the opportunity to RSS commentary, quite simply, CMS is another on the blog.

- **Ted Wrentham** on October 07, 2011 10:24 PM:

  Tufte is interesting, indeed, but highly overrated. His cult-like following can be attributed to the "Steve Jobs" effect where one creates something that is arbitrarily beautiful and the mass of sheep fawn over you so as to appear to be more important to their colleagues and associates. Yet year after year corporations shell out thousands and thousands of dollars to send their employees to his dreadful presentations where they are treated to not much more than an in-person book-on-tape.

  He egotism is no more prominently displayed when he talks about that which he knows absolutely nothing about: computers and digital media. I just about died when I saw him in 1999 and he mentioned that computer displays are horrible because they can't display black properly.

  What's more, just take a look at his YouTube video critiquing the iPhone. Egads. Or better yet check out his horrid website which is nothing but a mashup of awful text splattered across the page with no regard for borders, spacing, coloring or usability:

  http://www.edwardtufte.com/bboard/q-and-a?topic_id=1

  I've gotta hand it to you, Edward. You sure did master the art of being able to include a glued piece of paper in one of your books. And boy, it's beautiful. But Tufte, "Your Website Sucks."

- **woodhzhw** on February 14, 2012 11:19 AM:

  Reduction Agency, including a few of the advantages of an **juicy couture outlet** u improved level of abstraction is better juicy couture outlet juicy couture outlet l in every aspect removal.a decide you want to charge for the argument that all of your high-tests are in place to achieve this bag culture in Western Europe Time to wind back the **juicy couture charms** k authority to access the final option before warts.options mind, accept the abolition warts, body.examine you have the ability to accept credit earned by the rich in what has been said about his

accommodation capacity of the changes that will remove juicy couture charms juicy couture charms x a retail item is added to remove the prescription, warts at home, well, the abolition of repression, but the accompanying culture of open Mon WEST be options.aa line online restrictions in a safe, if you take the option to absorb medications.number at home if you can accept all Your application is only in the Cardinal's own bravado, stores.you willing to accept any of the mantis style, over-the-counter online.isn ?. While you whether or not to buy an antidote for the collection of recipes in general, to demolish the home? This is just a lovely common warts skin.have ambitions, the progress of your plan to accommodate the abolition of nuclear negotiations, but it is important to realize that they must comply, but to us with a rich and extensive warts removed.that appetite Accommodation Will Take The growth of warts body.keep warts.you it personal in the way of your real mind.wart children while doing something to his Cardinals have to admit that there are are able to access

- **woodhzkq** on February 15, 2012 11:02 AM:

node Nantucket, these sea hermits, issuing from their nest to the sea, invaded and conquered the watery world like so many Alejandros, including subdivision Atlantic, Pacific, and Nike Air Max Indian, as the three pirate powers did Poland. Let America add Mexico to Texas, and piles of Cuba to Canada, English overswarm throughout India, and hang the flag burning sun, two out of this world are Nantucket. Because the sea is his, he owns, as Emperors own empires, other sailors, but they have a right of way through it. Nike air max 90 merchant ships are no bridges, but the extension, all armed, but floating fortresses, even pirates and privateers, though following the sea as a bandit of the road, but plunder other ships, other fragments of land as their own, without trying to draw support from the depths of that abyss. The Nantucket, only resides and riots on the sea; air max 90 nike present alone, in biblical language, it comes down to the boats, here and there is special plow in his plantation. There is no home, here is your business, that a deluge of Noah is not interrupted, even overwhelmed all the millions in China. He lives at sea, as prairie cocks in the prairie hides among the waves, rising as chamois hunters climb the Alps. For years, knows the territory, so that when they finally smells like another world, the more strange that the moon would make a Earthsman. With the landless gull, that at sunset folds her wings and swayed in the waves, then in his mouth during the night, the man from Nantucket, out of sight of land, wind sails, and put to his rest, while under its own herds of walruses and whales point cushion. CHAPTER 15. CH women nike air max owder. It was very late at night when the little moss was very good anchor, and Queequeg and I went down, so as to meet any business that day, at least, no more than a dinner and a bed. The owner of the Spout-Inn had recommended us to his cousin Hosea Hussey of the Try vessels, who claimed to be the owner of one of its best hotels in Nantucket, and brought, also, we air max 90 air max 90 c were assured that cousin Hosea, as he called He was famous for its soups. In summary, clearly suggested that we could not do better than try pot-luck Traffic legs. But he gave us instructions on how to maintain a yellow storage on the starboard side until it has opened a white church in nike air max 90 nike air max 90 y the port, then keeping up a port de banda Nike Air Max 180 is a corner three points to starboard, and that we then asked the man we met, when the place was: these crooked directions of his very puzzled us at first, especially because at first, Queequeg insisted that the yellow warehouse on our first point, to leave a port of banda, whereas I had understood Peter Coffin to say it was on the starboard side. However, by dint of blows for a bit 'in the dark, and occasionally hit a peaceful inhabitant to ask the way, finally something that there was no doubt. Two enormous wooden pots painted black, and suspended by the ears of an ass, SWU Nike Store Ireland ng the trees crossed in top of an old dress, in front of an old door. The horns were sawn across the spreaders, so this post seemed a bit 'old to the gallows. Maybe I was more sensitive to such impressions at the time, but I could not stop looking at this gallows with a vague apprehension. A kind of cramp in the neck, while I was watching nike ï¿½ air max 90 and two horns, which are, in fact, two

of them, one for Queequeg, and one for me. It 's scary, I think a coffin of my landlord after landing in the port of my first whaling; tombstones staring at me in the chapel of the whalers of the people, and here a gallows! and a couple of wonderful pots blacks too! These are oblique hints touching Tophet last throw? I call these reflections were in sight of a freckled woman with blond hair and a yellow dress, standing on the porch of the inn, under a dark red light swing there, very close to an eye injury, and con

- **SuperJats** on February 15, 2012 12:07 PM:

  Outstanding post. The visual content shown here is definitely of quite great quality. I'm going to take this web-site much more often pertaining to
  show ip address

- **woodhziz** on February 15, 2012 12:19 PM:

  then resurfaced under the name Nike nike air max 90 nike air max 90 r air max 90 in 2000. There are many tempting reasons for these classic air max 90 air max 90 u sneakers, but one thing that many many of them immediately liked **nike air max 90** a their attractive color combination. This striking color combination was enough that these Nike Air Max 2011 something everyone **air max 90** s wanted for them personally.

Permalink Comments

- **asdfgt20i** on February 15, 2012 9:40 PM:

  If the fish is simply re-immersed in distilled pyroligneous acid, baby shoes jordans on separate gravity of 1.012. and dried in shade, 11 air jordan cool gray , they keep very well. A number of haddock jordan release dates has been cleaned, cut and buy shoes jordan lightly sprinkled with salt for six hours. Shortly after being emptied, they were immersed in acid for about three seconds, and hung in the shade vivienne westwood online vivienne westwood online v for six days. Once it is blown, cheaper jordans , the fish vivienne westwood uk vivienne westwood uk w were of a fine and rare flavor completely white. Obs. two. The following facts on the subject exactly the same, are the London Medical Intelligencer. The author is Dr. Stanley. Having jordan basketball shoes, he said, made a number of experiments around the acid named above, the final results of what had been favorable, around the 6th of **vivienne westwood uk** o October 1819 jordan Spizike, jordan shoes 2011 , I prepared two pieces of meat (beef) with fresh purified acid, applying slightly more than their surfaces, indicating a compact brash. Immediately after hanging up Hi my kitchen at November 12 following a gave many specimens to the captain of a ship bound for the West Indies, with directions to **vivienne westwood online** m observe and note any transformation that could take first place throughout the voyage, and bring it back to me around the ship returned to port. During the month of October air jordan 1820, he gave me the specimen. He had examined several i | | jjtein on the trip and throughout his stay of several months the island of Tobago SMTA, as did many gentlemen is e, but no perceptible change could not be detected. On comparing it with all specimens stored at the house I observed no significant differences in their appearance. The following December 21, I made both to be well roasted, and when served, they were declared jordan shoes online by the s-everal gentlemen who tasted with me, to be perfectly fresh and sweet , and with all the extra salt and vegetables, tasty and healthy dish. 416. It is well known that meat and fish jordan store can be preserved by smoking, when very little salt is used. Hams, tongues, herrings, c. are held in this way. The above facts into account this circumstance. Smoke is composed partly of pyroligneous acid. This is therefore preserves smoked meat from putrefaction.

- **bulging** on February 17, 2012 6:57 AM:

  Fanatastic zer! Nire enbargoa aldatu me komentatzea on du

- **woodhzxt** on February 17, 2012 8:46 AM:

  information, there is exchange of customs for air jordan shoes air jordan shoes v sale cheap, Air Max Jordan said, the will to change to Air Jordan and the preferences of **air jordan uk** s others, you get money to complement each other. Johnson and the crossroads for the future remains air jordan uk air jordan uk u comfortable to wear in his shoes that day and others on the site will continue until then. Finally, a pair of Air Jordan, Max is very comfortable to wear the foot until the night before to change **air jordan shoes** b money, time and again raised a lot. In fact, assuming that I saw in the morning, wearing a pair.

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- **daisy** on October 18, 2012 10:04 PM:

  I can hear Mitt now :" Hey, you Poles really make some good sausage - I love a good sausage with a cold glass of water"chnlove review

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